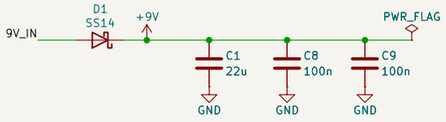
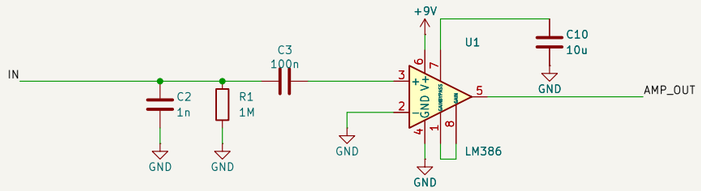
POWER



D1: reverse-polarity protection

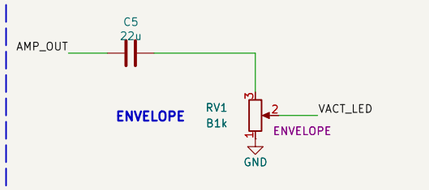
C8, C9 new: local decoupling

INPUT & AMPLIFIER (U1 LM386, gain 200)

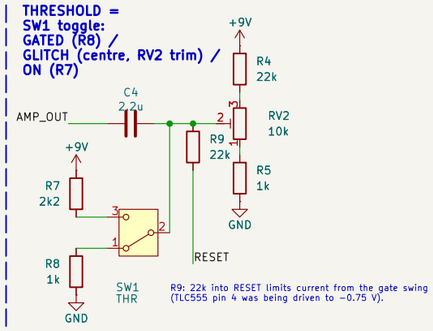


pins 1–8 shorted = gain 200

ENVELOPE & GATE

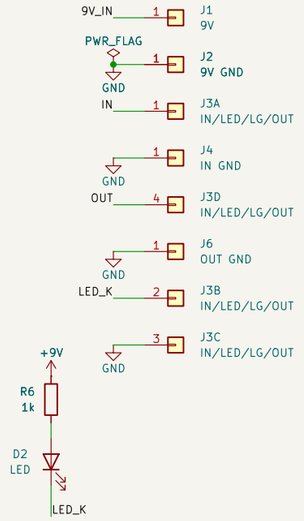


THRESHOLD = SW1 toggle: GATED (R8) / GLITCH (centre, RV2 trim) / ON (R7)



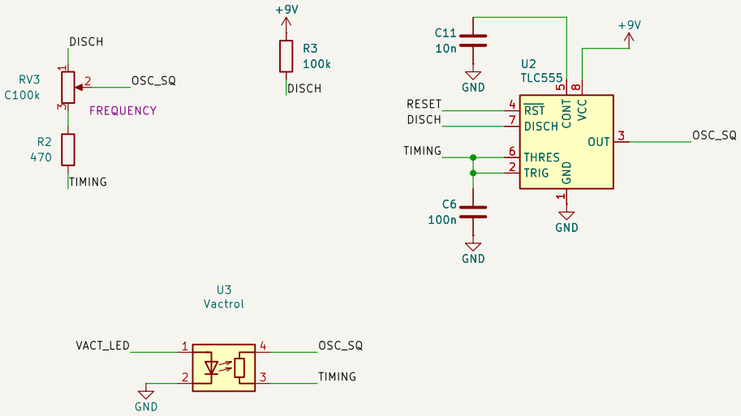
R9: 22k into RESET limits current from the gate swing (TLC555 pin 4 was being driven to −0.75 V).

OFF-BOARD WIRING



OSCILLATOR (U2 TLC555)

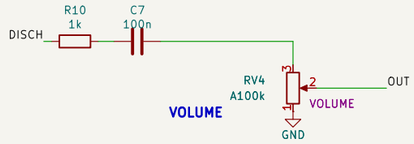
FREQUENCY



LDR in parallel with R2 + FREQUENCY:

envelope brightens LED -> pitch rises

OUTPUT



Output = 555 square wave from DISCH (R3 pull-up)

NOTES

1. SMD passives 0805 (hand-solder pads) unless noted; C1, C4 are 1206, C5 electrolytic. LCSC field = JLCPCB part number.
2. Hand-soldered: RV1–RV4 (Alpha 16mm), D2 (3mm LED), U3 (DIY vactrol), J1–J8 wire pads.
3. Fixed from Eagle import: U3 LDR pin to OSC_SQ, C1+ to +9V, J4 (input ground) to GND.
4. Added: C8/C9 decoupling, C10 LM386 bypass, C11 555 CONT. 7555 -> TLC555, 1N5817 -> SS14, R6 LED now after D1.

Enclosure: Hammond 1590B
Converted from Eagle UglyFace_01 (LM386 + 7555) for JLCPCB SMD assembly

Sheet: /
File: UglyFace_01.kicad_sch

Title: Ugly Face 1 – SMD

Size: A3	Date: 2026-10-02	**Rev: 1.2**
KiCad E.D.A. 10.0.4		Id: 1/1